

ANSWER KEY

GATE EC — GATE EC 2026 Trend-Based Mock Paper

4

Q.No.	Type	Answer	Marks
1	2MCQ	30 years	2
2	3MCQ	B	3
3	2MCQ	B	2
4	2MCQ	A	2
5	2MCQ	B	2
6	2NAT	4	2
7	3MCQ	C	3
8	2MCQ	C	2
9	2MSQ	A,B	2
10	2MSQ	A	2
11	2MCQ	B	2
12	2MCQ	could have been	2
13	2NAT	B	2
14	1NAT	B	1
15	1NAT	B	1
16	2MCQ	2	2
17	2NAT	0.1	2
18	1NAT	A	1
19	2MCQ	C	2
20	2MCQ	$AB + AC + BC$	2
21	1MCQ	C	1
22	2NAT	0.001	2
23	1MCQ	B	1
24	1MCQ	B	1
25	1MCQ	B	1
26	1MCQ	B	1
27	1MCQ	B	1
28	1NAT	C	1
29	1MCQ	C	1
30	1NAT	B	1
31	1MCQ	A	1
32	1MCQ	A	1
33	1MCQ	B	1
34	1MCQ	A	1
35	1MCQ	A	1
36	1NAT	A	1

37	1MCQ	B	1
38	1MCQ	A	1
39	1NAT	B	1
40	1MCQ	A	1
41	1NAT	A	1
42	1MCQ	A	1
43	1MCQ	A	1
44	1MCQ	C	1
45	1MCQ	A	1
46	1MCQ	B	1
47	1MCQ	D	1
48	2MCQ	A,B	2
49	2MCQ	A,C	2
50	2MCQ	B	2
51	1MCQ	A	1
52	2NAT	2×10^{20}	2
53	2MCQ	A	2
54	2MCQ	B	2
55	1MCQ	B	1
56	2MCQ	A	2
57	2MCQ	2	2
58	2NAT	0.5	2
59	2NAT	3	2
60	2MCQ	49.92 dB	2
61	2MCQ	1	2
62	2MCQ	A	2
63	2NAT	0.00	2
64	1MCQ	B	1
65	2MSQ	A,C	2

Q1. General Aptitude | 2 marks | 2MCQ

Five years ago, the ratio of Aman's age to his father's age was 1:4, and five years from now, the ratio will be 2:5. What...

Answer: 30 years

Let A and F be Aman's and his father's present ages. Five years ago $(A-5)/(F-5)=1/4 \Rightarrow F=4A-15$. Five years later $(A+5)/(F+5)=2/5 \Rightarrow F=(5A+15)/2$. Equating gives $4A-15=(5A+15)/2 \Rightarrow A=15, F=45$. Father's age when Aman was born = $F - A = 45 - 15 = 30$ years.

Source: Adapted from GATE 2024 EC Session 7 Q3

Q2. General Aptitude | 3 marks | 3MCQ

Which one of the following plots represents $f(x) = -a|x|$, where a is a non-zero real number? Note: The figures shown are rep...

Answer: B

$f(x) = -a|x|$ ($a \neq 0$) is a V shaped graph with its vertex at the origin and opening downwards (if $a > 0$) or upwards (if $a < 0$). The only plot that matches the characteristic inverted V shape is option B.

Source: Adapted from GATE 2025 EC Session 1 Q5

Q3. General Aptitude | 2 marks | 2MCQ

Sum of first 100 natural numbers:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q10

Q4. General Aptitude | 2 marks | 2MCQ

Choose opposite of 'benevolent':

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q10

Q5. General Aptitude | 2 marks | 2MCQ

For a real number $x > 1$, $\frac{1}{x} + \log_2 x + \log_3 x + \log_4 x = 1$. The value of x is

Answer: B

$\log_2 x + \log_3 x + \log_4 x = 1 - \frac{1}{x}$. Solving; $x = 2$. Option B.

Source: Adapted from GATE 2024 EC Session 7 Q4

Q6. General Aptitude | 2 marks | 2NAT

P, Q, R, S, and T have launched a new startup. Two of them are siblings. The office of the startup has just three rooms....

Answer: 4

Startup room assignment problem. Counting valid configurations gives 4.

Source: Adapted from GATE 2024 EC Session 7 Q2

Q7. General Aptitude | 3 marks | 3MCQ

The greatest prime factor of $(3199 - 3196)$ is

Answer: C

$3199 - 3196 = 3$, and the only prime factor of 3 is 3 itself, which is the greatest prime factor.

Source: Adapted from GATE 2024 EC Session 7 Q5

Q8. General Aptitude | 2 marks | 2MCQ

If '*b*' denotes increasing order of intensity, then the meaning of the words [charm enamor bewitch] is analogous to [bored...]

Answer: C

Charm < enamor < bewitch. Bored < intrigued < fascinated. 'Intrigued' fits.

Source: Adapted from GATE 2024 EC Session 7 Q1

Q9. General Aptitude | 2 marks | 2MSQ

The following figures show three curves generated using an iterative algorithm. The total length of the curve generated ...

Answer: A,B

Curve generation algorithm. Analysis shows curves matching options A and B.

Source: Adapted from GATE 2025 EC Session 1 Q4

Q10. General Aptitude | 2 marks | 2MSQ

The 12 musical notes are given as $\sharp, \sharp, \flat, \flat, \sharp, \flat, \sharp, \flat, \sharp, \flat, \sharp, \flat$. Frequency of each note is 122 times the frequency of ...

Answer: A

12 musical notes frequency pattern. Analysis gives option A.

Source: Adapted from GATE 2025 EC Session 1 Q3

Q11. General Aptitude | 2 marks | 2MCQ

Choose synonym of 'trivial':

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q10

Q12. General Aptitude | 2 marks | 2MCQ

Had I learnt acting as a child, I _____ a famous film star. Select the most appropriate option to complete the above ...

Answer: could have been

The clause 'Had I learnt acting as a child' is a past unreal condition, requiring the result clause to use the perfect conditional form 'could have been' to indicate a hypothetical outcome in the past.

Source: Adapted from GATE 2025 EC Session 1 Q2

Q13. Engineering Mathematics | 2 marks | 2NAT

Integral of $1/x$ dx:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q1

Q14. Engineering Mathematics | 1 marks | 1NAT

$i^4 =$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q1

Q15. Engineering Mathematics | 1 marks | 1NAT

$P(A \cap B) = ?$

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q1

Q16. Network, Signals & Systems | 2 marks | 2MCQ
In the given circuit, the current I_x (in mA) is _____. 1000 10 -+ 5 mA 1k 1k 1k 10 I_x 2 mA

Answer: 2

Circuit analysis with given currents and resistors. KCL/KVL gives $I_x = 2$ mA.
Source: Adapted from GATE 2024 EC Session 1 Q31

Q17. Network, Signals & Systems | 2 marks | 2NAT
Two resistors are connected in a circuit loop of area 5 m², as shown in the figure below. The circuit loop is placed on ...

Answer: 0.1

Circuit loop in magnetic field. Faraday's law gives induced emf = 0.1V.
Source: Adapted from GATE 2025 EC Session 1 Q65

Q18. Network, Signals & Systems | 1 marks | 1NAT
Z-parameter conditions:

Answer: A

Correct Answer: A
Source: Adapted from GATE 2021 EC Session 1 Q8

Q19. Network, Signals & Systems | 2 marks | 2MCQ
Let U and V denote the set of real numbers and the three dimensional vector space over it, respectively. The value of $\dim(U \otimes V)$ for ...

Answer: C

Vector space operations. Inner product/norm calculation gives option C.
Source: Adapted from GATE 2024 EC Session 1 Q30

Q20. Network, Signals & Systems | 2 marks | 2MCQ
A 3-input majority logic gate has inputs A , B , and C . The output of the gate is logic '1' if two or more of the inputs are ...

Answer: AB + AC + BC

The majority function is true when at least two inputs are 1, which occurs for the three possible pairs of inputs being 1 simultaneously. Hence the Boolean expression is the sum of the three pairwise AND terms: $AB + AC + BC$.
Source: Adapted from GATE 2025 EC Session 1 Q23

Q21. Network, Signals & Systems | 1 marks | 1MCQ
 h -parameter represents:

Answer: C

Correct Answer: C
Source: Adapted from GATE 2023 EC Session 1 Q7

Q22. Network, Signals & Systems | 2 marks | 2NAT
An electric field of 0.01 V/m is applied along the length of a copper wire of circular cross-section with diameter 1 mm....

Answer: 0.001

Copper wire in electric field. Current density and conductivity give answer 0.001 A.
Source: Adapted from GATE 2025 EC Session 1 Q50

Q23. Electronic Devices | 1 marks | 1MCQ
 n -type majority carriers:

Answer: B

Correct Answer: B
Source: Adapted from GATE 2024 EC Session 1 Q3

Q24. Electronic Devices | 1 marks | 1MCQ
MOSFET is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 EC Session 1 Q8

Q25. Electronic Devices | 1 marks | 1MCQ
SCR has:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q9

Q26. Electronic Devices | 1 marks | 1MCQ
Zener diode used for:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 EC Session 1 Q9

Q27. Analog Circuits | 1 marks | 1MCQ
Common emitter has:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q4

Q28. Analog Circuits | 1 marks | 1NAT
; —â \$¥B —3

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 EC Session 1 Q3

Q29. Analog Circuits | 1 marks | 1MCQ
Ideal op-amp open-loop gain:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2023 EC Session 1 Q2

Q30. Analog Circuits | 1 marks | 1NAT
FET is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q3

Q31. Analog Circuits | 1 marks | 1MCQ
RC oscillator frequency:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q9

Q32. Digital Circuits | 1 marks | 1MCQ
4-bit binary counter counts:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q4

Q33. Digital Circuits | 1 marks | 1MCQ
D flip-flop follows input after:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q5

Q34. Digital Circuits | 1 marks | 1MCQ
ROM is:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 EC Session 1 Q4

Q35. Digital Circuits | 1 marks | 1MCQ
TTL stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q3

Q36. Control Systems | 1 marks | 1NAT
Routh-Hurwitz determines:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2024 EC Session 1 Q7

Q37. Control Systems | 1 marks | 1MCQ
Lead compensator:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q6

Q38. Control Systems | 1 marks | 1MCQ
Nyquist criterion uses:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q6

Q39. Control Systems | 1 marks | 1NAT
Transfer function is defined:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 EC Session 1 Q5

Q40. Communication Systems | 1 marks | 1MCQ
AM stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2024 EC Session 1 Q6

Q41. Communication Systems | 1 marks | 1NAT
PCM stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 EC Session 1 Q4

Q42. Communication Systems | 1 marks | 1MCQ
FM has:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 EC Session 1 Q5

Q43. Communication Systems | 1 marks | 1MCQ
ASK modulation varies:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q5

Q44. Electromagnetics | 1 marks | 1MCQ
Maxwell has how many equations:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 EC Session 1 Q8

Q45. Electromagnetics | 1 marks | 1MCQ
Characteristic impedance Z_0 =:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q7

Q46. Electromagnetics | 1 marks | 1MCQ
Waveguide supports:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q7

Q47. Electromagnetics | 1 marks | 1MCQ
Skin depth increases with:

Answer: D

Correct Answer: D

Source: Adapted from GATE 2023 EC Session 1 Q6

Q48. Network, Signals & Systems | 2 marks | 2MCQ
In the circuit below, the opamp is ideal. $R_1 = C$, $R_2 = R$, $R_3 = R$, $R_4 = R$. If the circuit is to show sustained oscillations, the respect...

Answer: A,B

Opamp oscillator conditions. Barkhausen criterion matches options A and B.

Source: Adapted from GATE 2024 EC Session 1 Q50

Q49. Network, Signals & Systems | 2 marks | 2MCQ
Which of the following statements is/are TRUE with respect to ideal MOSFET-based DC-coupled single-stage amplifiers hav...

Answer: A,C

MOSFET amplifier analysis. Statements about DC-coupled single-stage amplifiers. Options A and C are true.

Source: Adapted from GATE 2025 EC Session 1 Q30

Q50. Network, Signals & Systems | 2 marks | 2MCQ
For a causal discrete-time LTI system with transfer function $H(z) = (2z^2 + 3z + 13) / (z^2 - 13z + 13)$ - which of the following statements ...

Answer: B

Discrete-time LTI system. Stability and frequency response analysis gives option B.

Source: Adapted from GATE 2024 EC Session 1 Q24

Q51. Network, Signals & Systems | 1 marks | 1MCQ
FS exists for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 EC Session 1 Q2

Q52. Network, Signals & Systems | 2 marks | 2NAT
The free electron concentration profile $n(x)$ in a doped semiconductor at equilibrium is shown in the figure, where the p ...

Answer: 2×10^{20}

Semiconductor carrier concentration profile. Integration/gradient analysis gives $n = 2 \times 10^{20} / \text{m}^3$.

Source: Adapted from GATE 2024 EC Session 1 Q26

Q53. Network, Signals & Systems | 2 marks | 2MCQ
Consider a system S represented in state space as $\dot{x} = 0 - 2x + 1r$, $y = 2 - 5x$. $\det(1 - 3s)$ Which of the state space rep...

Answer: A

State space representation. Controllability/observability analysis gives option A.

Source: Adapted from GATE 2024 EC Session 1 Q53

Q54. Network, Signals & Systems | 2 marks | 2MCQ
Two fair dice (with faces labeled 1, 2, 3, 4, 5, and 6) are rolled. Let the random variable denote the sum of the outcom...

Answer: B

Two dice sum. Probability distribution analysis gives option B.

Source: Adapted from GATE 2025 EC Session 1 Q58

Q55. Electronic Devices | 1 marks | 1MCQ
 n -type majority carriers:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 EC Session 1 Q3

Q56. Network, Signals & Systems | 2 marks | 2MCQ

The general form of the complementary function of a differential equation is given by $y(t) = (At + B)e^{-2t}$, where A and B are constants.

Answer: A

Differential equation complementary function. Analysis shows form matching option A.

Source: Adapted from GATE 2024 EC Session 1 Q11

Q57. Network, Signals & Systems | 2 marks | 2MCQ

An amplitude modulator has output (in Volts) $s(t) = A\cos(400t) + B\cos(360t) + B\cos(440t)$. The carrier power normalized to the average power is

Answer: 2

AM modulator output $s(t) = A\cos(400t) + B\cos(360t) + B\cos(440t)$. Carrier power normalization gives answer 2.

Source: Adapted from GATE 2024 EC Session 1 Q28

Q58. Network, Signals & Systems | 2 marks | 2NAT

In the circuit given below, the switch S was kept open for a sufficiently long time and is closed at time $t = 0$. The time constant of the circuit is

Answer: 0.5

RC circuit transient. Time constant and response analysis gives answer 0.5.

Source: Adapted from GATE 2024 EC Session 1 Q32

Q59. Network, Signals & Systems | 2 marks | 2NAT

For a causal discrete-time LTI system with transfer function $H(z) = \frac{1}{(z - \frac{1}{2})(z - \frac{1}{3})}$, which of the following statements is/are true?

Answer: 3

Discrete-time system analysis. Stability/frequency response gives answer 3.

Source: Adapted from GATE 2024 EC Session 1 Q24

Q60. Network, Signals & Systems | 2 marks | 2MCQ

Let $X(t) = A\cos(2\pi f_0 t + \phi)$ be a random process, where amplitude A and phase ϕ are independent of each other, and are uniformly distributed over $[0, 2\pi]$ and $[-\pi, \pi]$ respectively. The power spectral density of X(t) is

Answer: 49.92 dB

The signal range is $[-2, 2]$ so step size $\Delta = 2/9 = 0.222$. Quantization noise variance $= \frac{\Delta^2}{12} = 0.004938$. Signal variance from given autocorrelation is $\frac{1}{2} \int_{-\infty}^{\infty} R_X(\tau) d\tau = 1$. Hence $SNR = \frac{\text{Signal variance}}{\text{Quantization noise variance}} = \frac{1}{0.004938} = 202.5$. $10 \log_{10}(202.5) = 23.06$ dB.

Source: Adapted from GATE 2024 EC Session 1 Q57

Q61. Network, Signals & Systems | 2 marks | 2MCQ

As shown in the circuit, the initial voltage across the capacitor is 10 V, with the switch being open. The switch is the...

Answer: 1

RC circuit with switch. Transient analysis gives answer 1.

Source: Adapted from GATE 2024 EC Session 1 Q35

Q62. Network, Signals & Systems | 2 marks | 2MCQ

The opamps in the circuit shown are ideal, but have saturation voltages of ± 10 V. Assume that the initial inductor current is 0 A.

Answer: A

Opamp circuit with saturation. Analysis of inductor current and output gives option A.

Source: Adapted from GATE 2024 EC Session 1 Q49

Q63. Network, Signals & Systems | 2 marks | 2NAT

Let X and Y be independent Bernoulli random variables taking values in $\{0, 1\}$. The joint probability mass function of the random variables is

Answer: 0.00

The marginals are $p_X(0)=0.20$, $p_X(1)=0.80$ and $p_Y(0)=0.30$, $p_Y(1)=0.70$. Each joint probability equals the product of the corresponding marginals (e.g., $0.06 = 0.20 \times 0.30$, $0.14 = 0.20 \times 0.70$, etc.), indicating X and Y are independent. Hence the mutual information $I(X;Y) = -\sum p(x,y) \log_2 \frac{p(x,y)}{p_X(x)p_Y(y)} = 0$, which rounded to two decimal places is 0.00.

Source: Adapted from GATE 2025 EC Session 1 Q60

Q64. Control Systems | 1 marks | 1MCQ
Lead compensator:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 EC Session 1 Q6

Q65. Network, Signals & Systems | 2 marks | 2MSQ
Consider a non-negative function $f(t)$ which is continuous and bounded over the interval $[2, 8]$. Let $F(s)$ and $f(t)$ denote, respectively...

Answer: A,C

Continuous function on $[2,8]$. Calculus properties (mean value, integration). Options A and C are true.

Source: Adapted from GATE 2025 EC Session 1 Q51